

Fast Realistic Rendering

SCREEN-SPACE AMBIENT OCCLUSION

PROJECT STATEMENT

In this assignment, we will deal with the lighting approximation based on the environment occlusion. The goal is to get familiar with the elements that are under the hood of Ambient Occlusion (AO) in real time. To do so, you have to implement a method to estimate AO in image space.

There are several methods in the literature that estimates ambient occlusion in image space in real-time:

- Louis Bavoil, Miguel Sainz, and Rouslan Dimitrov. "Image-space Horizon- based Ambient Occlusion". In: *ACM SIGGRAPH 2008 Talks*. SIGGRAPH '08. Los Angeles, California: ACM, 2008, 22:1–22:1. ISBN: 978-1-60558-343- 3. DOI: 10.1145/1401032.1401061. URL: <http://doi.acm.org/10.1145/1401032.1401061>.
- Martin Mittring. "Finding Next Gen: CryEngine 2". In: *ACM SIGGRAPH 2007 Courses*. SIGGRAPH '07. San Diego, California: ACM, 2007, pp. 97– 121. ISBN: 978-1-4503-1823-5. DOI: 10.1145/1281500.1281671. URL: <http://doi.acm.org/10.1145/1281500.1281671>.
- Dominic Fillion and Rob McNaughton. "Effects & Techniques". In: *ACM SIGGRAPH 2008 Games*. SIGGRAPH '08. Los Angeles, California: ACM, 2008, pp. 133–164. DOI: 10.1145/1404435.1404441. URL: <http://doi.acm.org/10.1145/1404435.1404441>
- Morgan McGuire et al. "The Alchemy Screen-Space Ambient Obscurance Algorithm". In: *High-Performance Graphics 2011*. Vancouver, BC, Canada, 2011. URL: <https://dl.acm.org/doi/abs/10.1145/2018323.2018327>

In this project, we ask you to read these techniques and **implement the first one** of them. You can extend your PBR project, or you can start again using the given skeleton of the first assignment.

Particularly, we ask you to:

1. Implement a two-step renderer of the scene that can show the albedo, the normal, and the scene's depth. You may include a GUI element to allow the user to choose between them.
2. Implement (using shaders) the SSAO technique. You should use a basic kernel equal for all the pixels. Modify the GUI to allow the user to enable/disable the visualization of AO estimation.
3. Improve your implementation to reduce the present artifacts by randomizing the samples and including the third rendering step to blur your estimation. Modify the GUI to allow the user to enable/disable these improvements.
4. Combine AO with the albedo of the model. If it is available, you can also combine it with your PBR result. Modify the GUI to allow the user to show your final result.

DELIVERY

Please upload a single zip file **by June 2nd**, named using “SSAO” and your username. For instance: SSAO-imanol.munoz.zip.

The zip file should contain:

- A compilable and executable project. This includes:
 - All the required .c, .cc, .cpp, .h, .hpp, .ui, etc. files needed to compile your application (including the map generator).
 - All the code used by your shaders.
 - All the used texture files.
 - A .PRO file, Makefile, or similar script that can compile and generate an executable file out of your source files.
 - A list of the dependencies needed to compile your application.
- A short report explaining the implemented solution in PDF format, following the same template of the previous assignment. The report may include:
 - A general explanation of the technique. You can use draws to illustrate it. You may also include a short theoretical comparison of this technique concerning the other mentioned ones.
 - A very well-documented version of your shaders. It should be clear which lines of the code relate to the different parts of the algorithms.
 - A discussion of your project where you describe the problems found and the proposed solutions. You can use images/screenshots to illustrate them.
- A video presentation of your project (it can be a link to a cloud service like Youtube). This video presentation will include a demonstration execution of the functionalities of your code described by you with a voice-over. You need to discuss results, limitations, and anything you find relevant for your grade.

Remember:

You can find several examples of how to implement this kind of algorithm on the Internet. **Any code directly copied from or delivering a mimetic solution (without a fair justification) to existing examples will imply failing the assignment.**